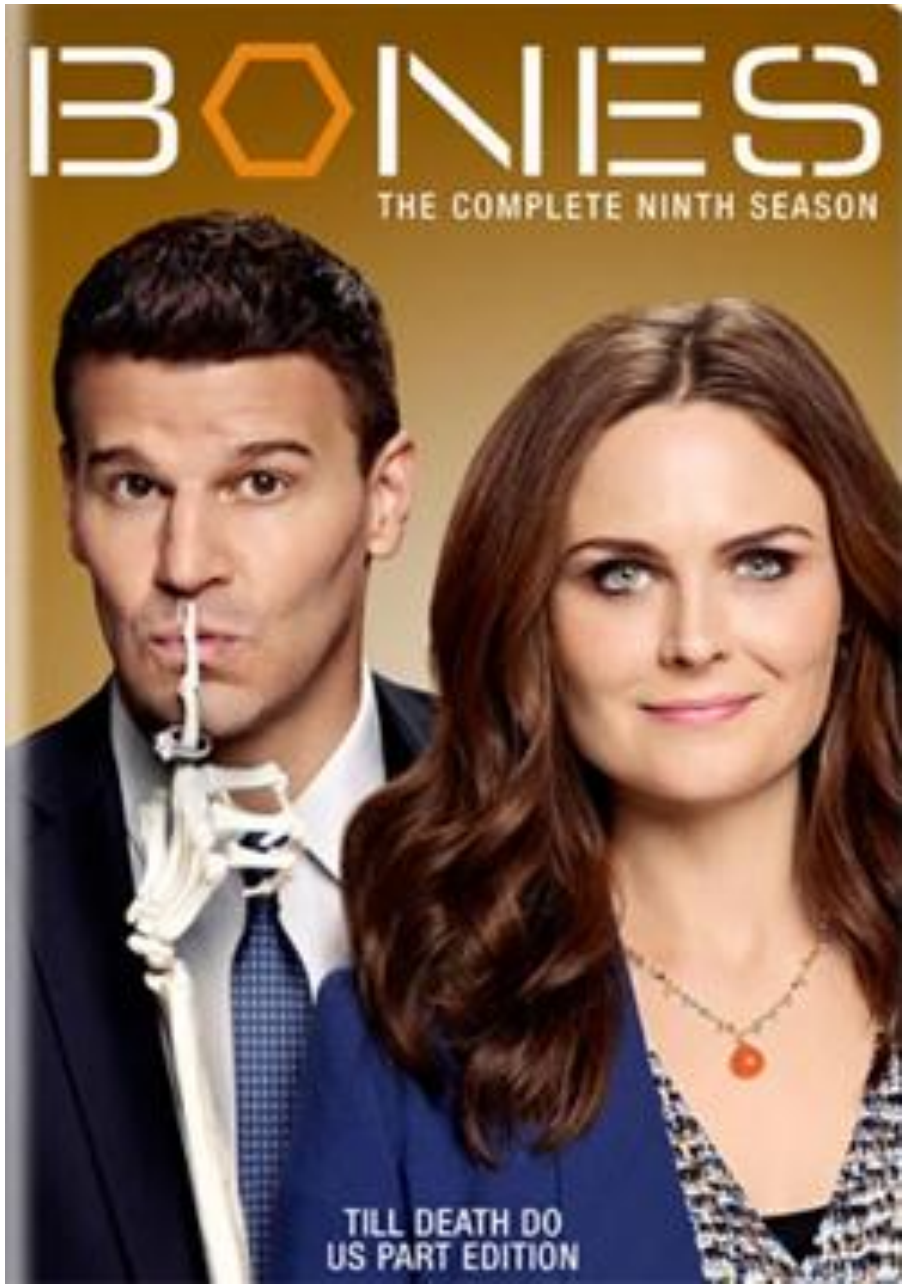
Determining Bone Health Utilizing Visible Light

Sophie Krajmalnik
Medicine and Health

Inspiration



Rationale/ Initial Hypothesis

If a bone sample is exposed to a controlled source of light,

then the amount of light transmitted through the bone will differ between healthy bone and porous bone.

Background Information

Bone

- Collagen provides flexibility and calcium adds rigidity
- Compact/cortical and spongy

Osteoporosis and T-score

- Increased porosity and increased risk of fracture
- Unit for measuring bone density
- Healthy bone: -1 to 1 and osteoporosis diagnosis: -2.5

Bone Sample Preparation

- Durham University study
- Boiling a bone increases crystallinity and porosity and decreases protein and mineral content

Engineering Goals

Design and prototype a device that can measure the difference in bone porosity between healthy bone and porous bone using visible light, measured using a Light Dependent Resistor.



Cost-Effective

Quick and Easy



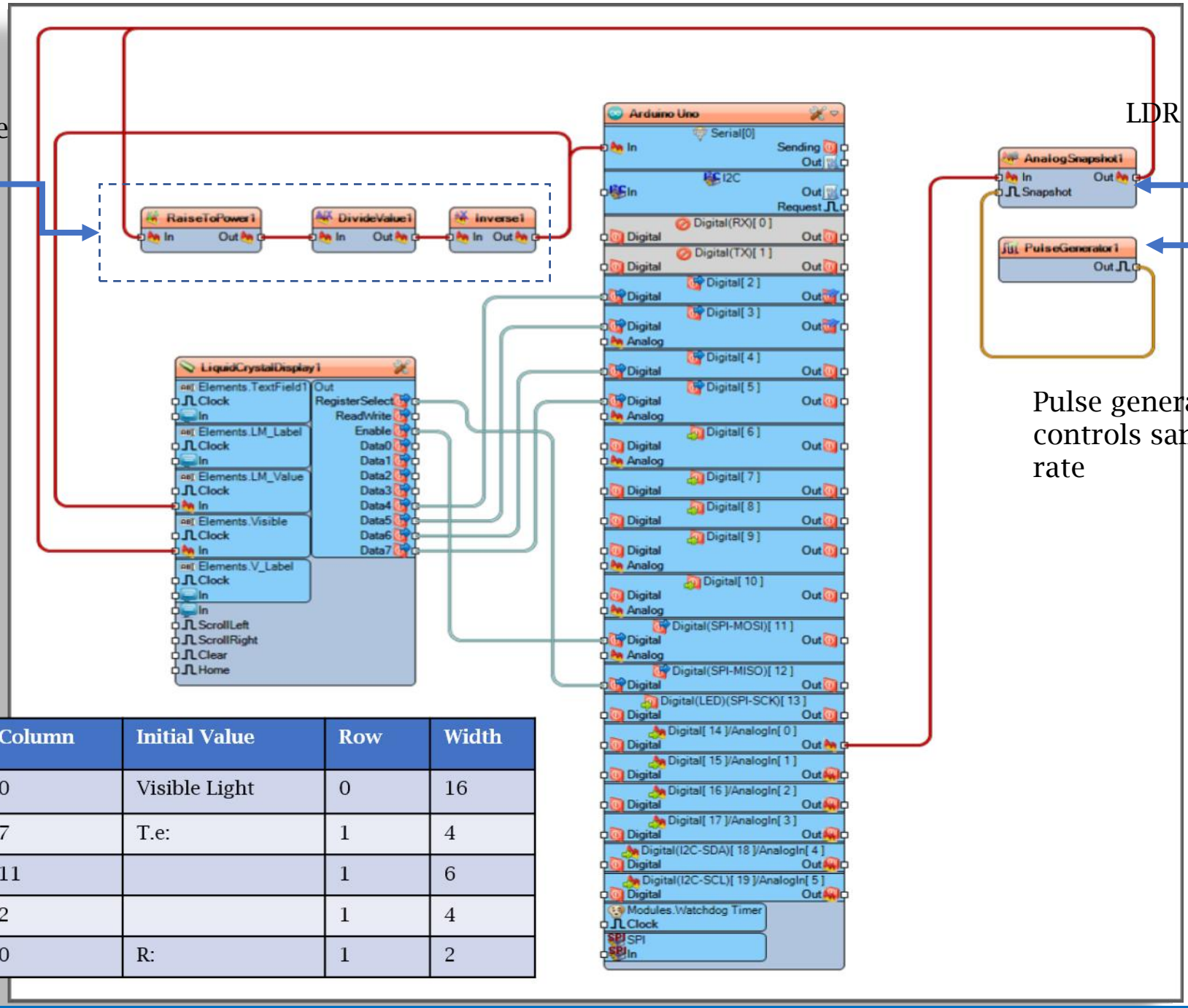
Portable



Safe

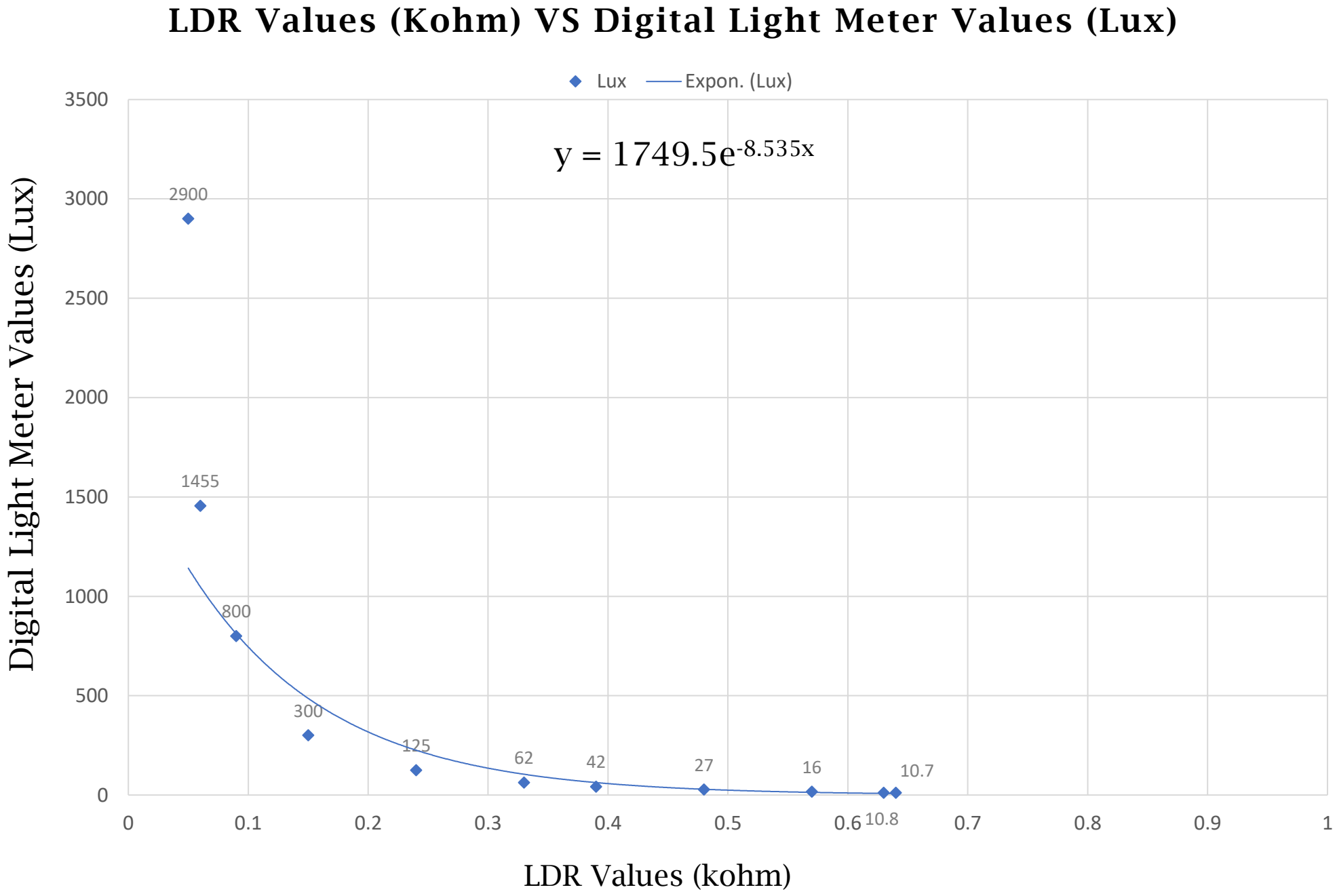
Logical Design

Mock T-Score Calculation

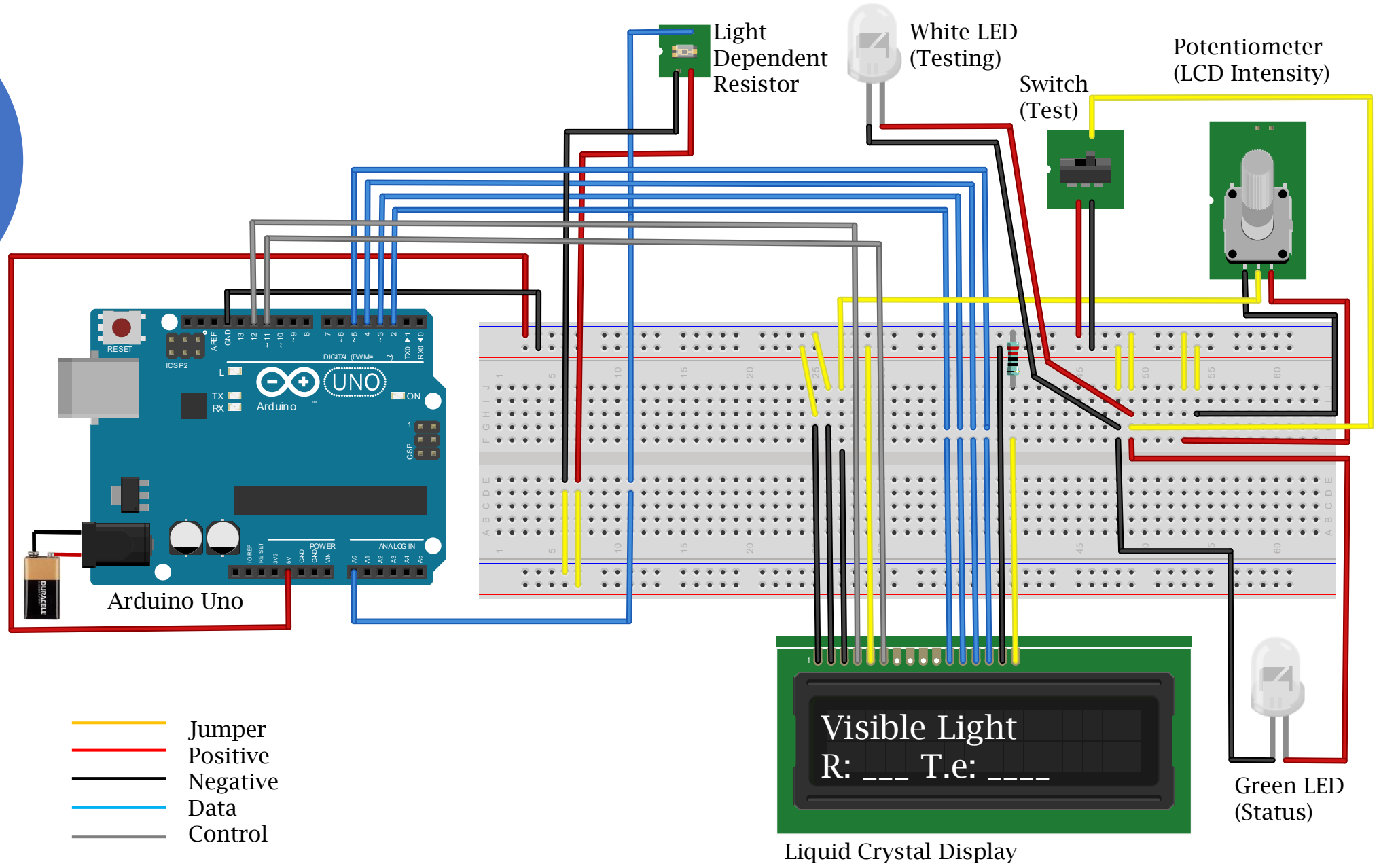


LCD Element	Column	Initial Value	Row	Width
Text Field 1	0	Visible Light	0	16
LM_Label	7	T.e:	1	4
LM_Value	11		1	6
Visible	2		1	4
V_Label	0	R:	1	2

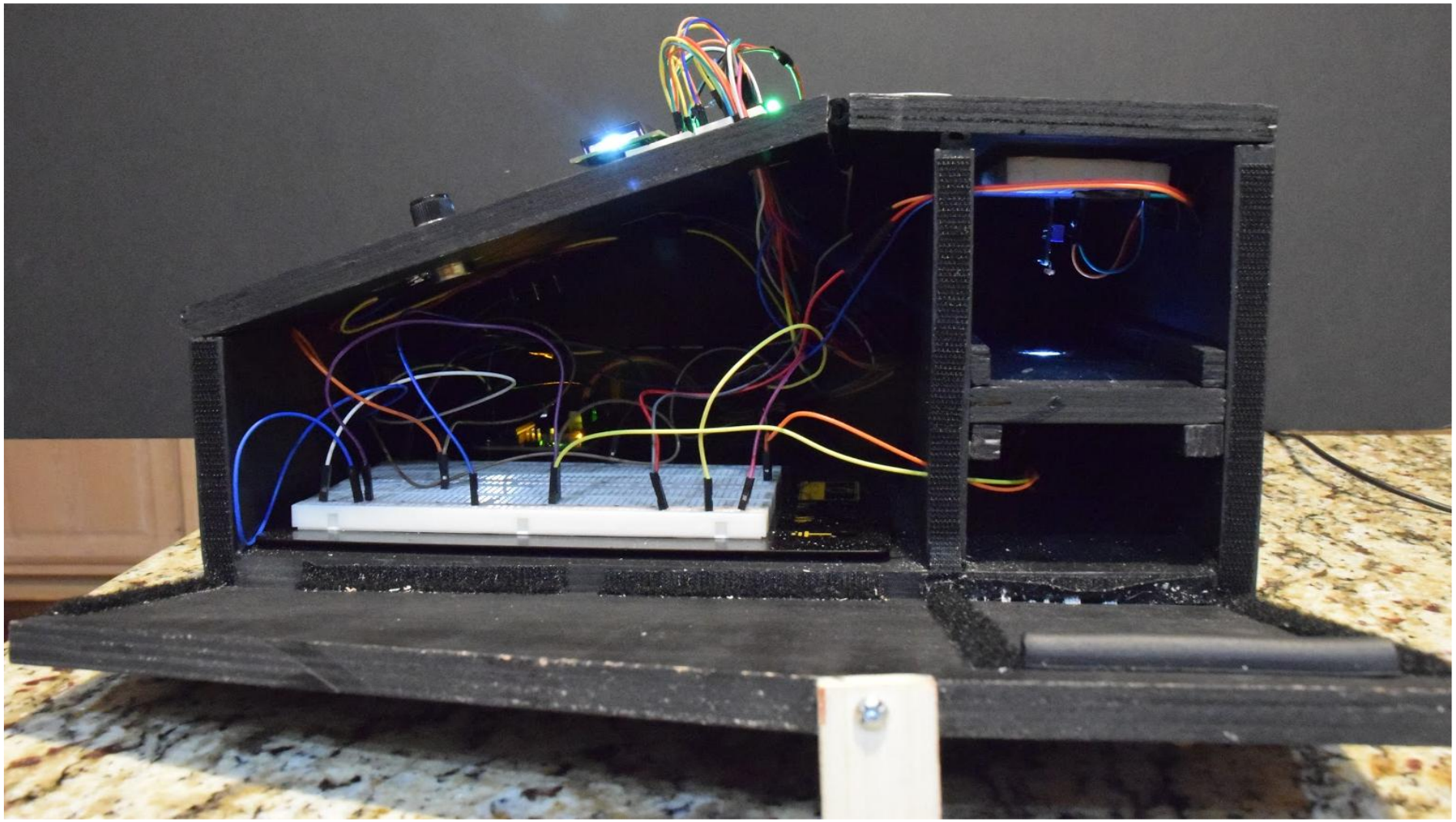
Sensor Calibration



Prototype Schematic



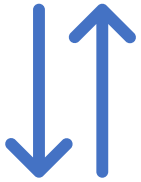
Prototype



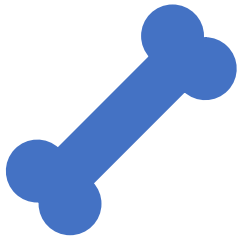
Hypotheses



If a bone sample is exposed to a controlled source of light, then the amount of light transmitted through the bone will differ between healthy bone and porous bone.

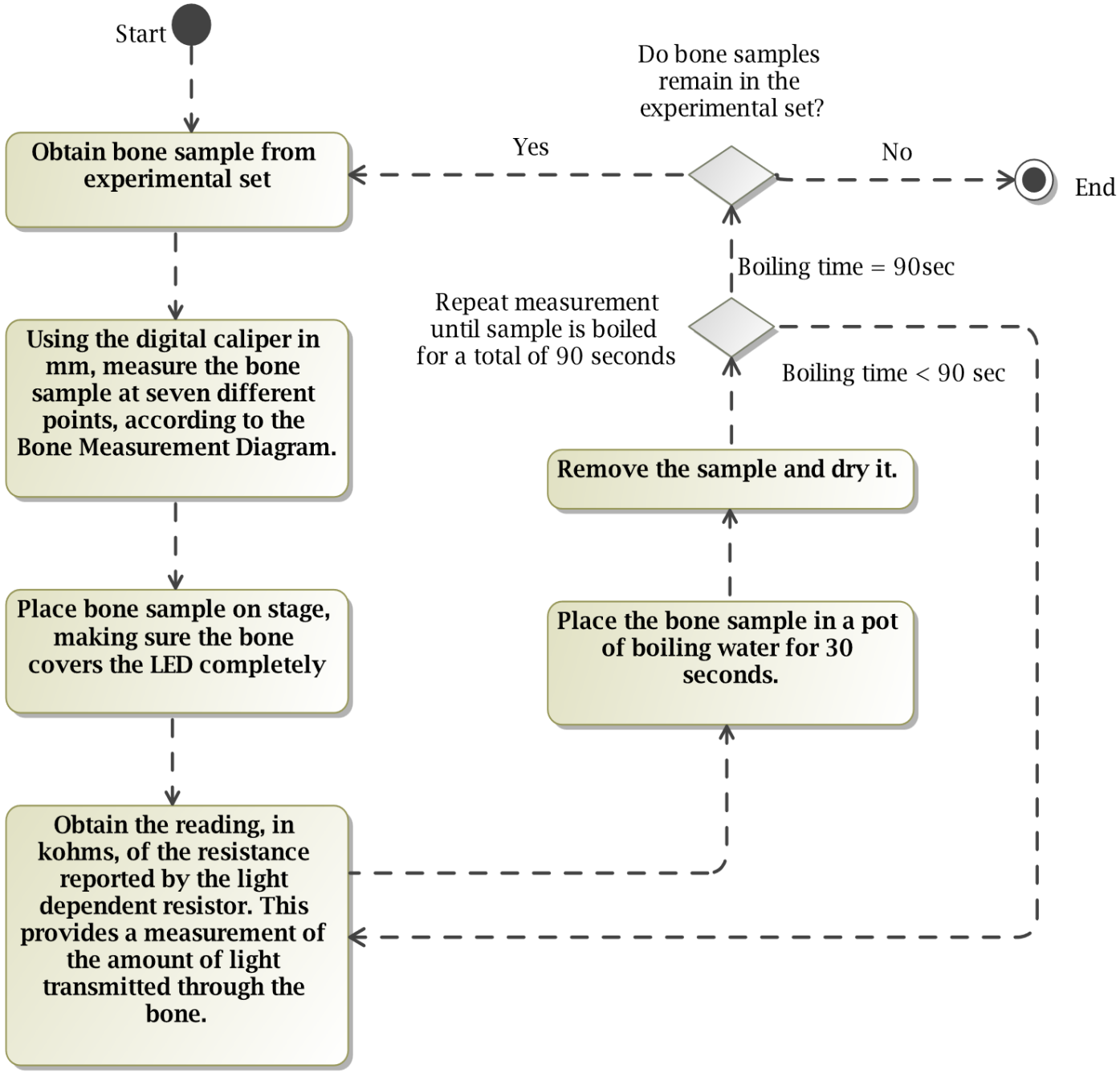


If porosity increases in a bone sample, then translucency will decrease.

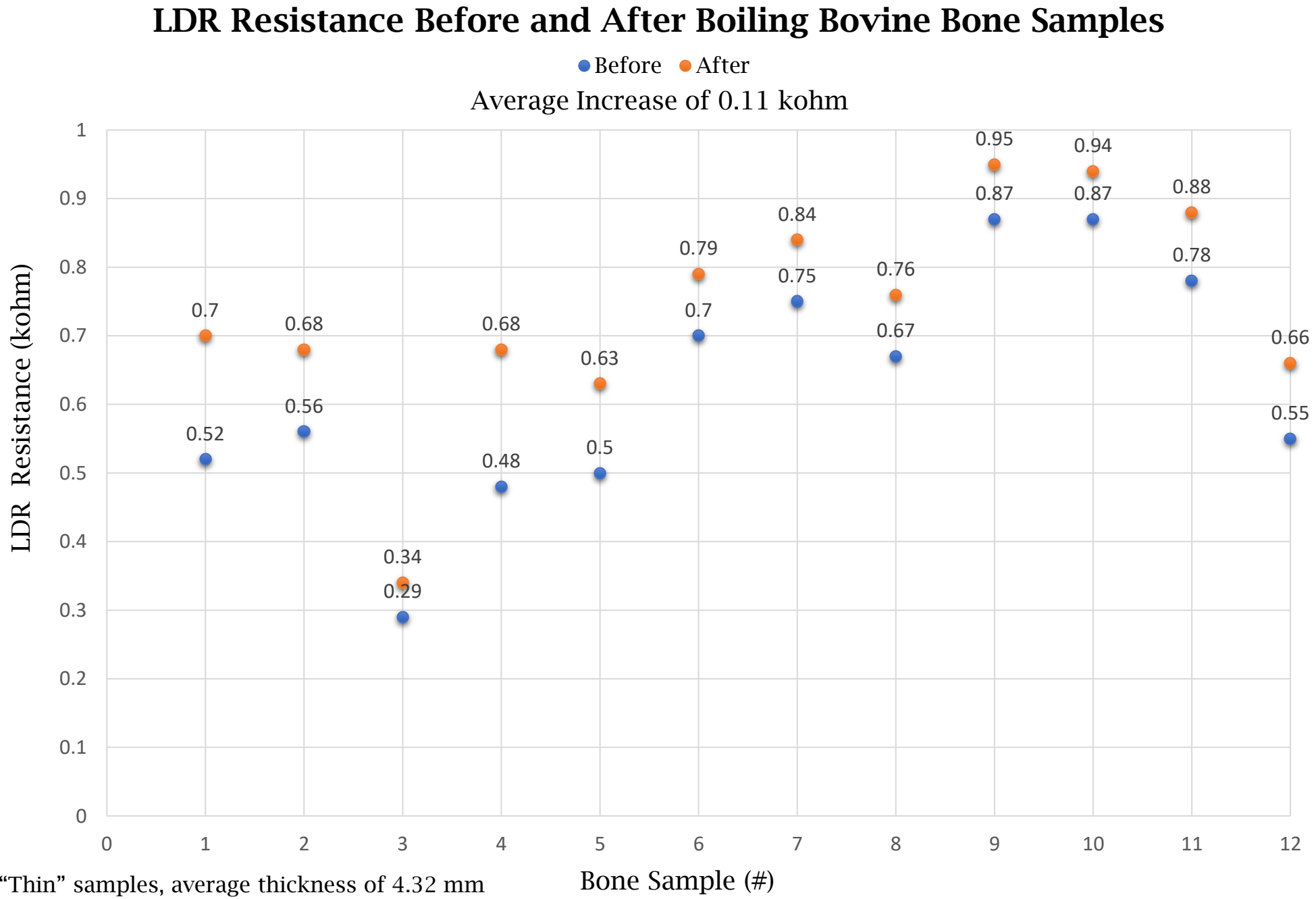


If human bone is submitted to the same testing as bovine bone, then the human bone will have different exact data points from bovine bone but show a similar trend.

Procedure

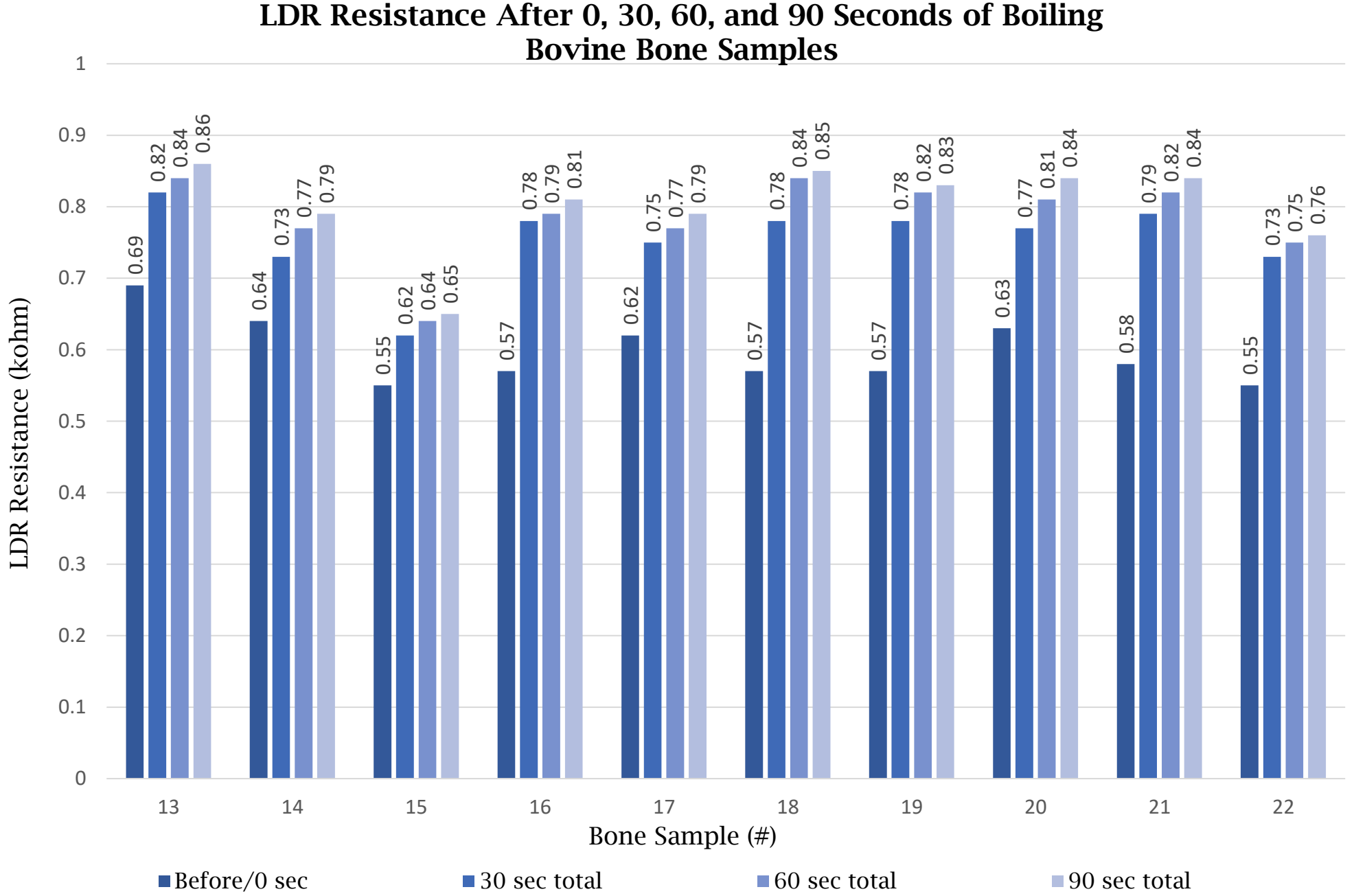

Start
Obtain bone sample from experimental set
Using the digital caliper in mm, measure the bone sample at seven different points, according to the Bone Measurement Diagram.
Place bone sample on stage, making sure the bone covers the LED completely
Obtain the reading, in kohms, of the resistance reported by the light dependent resistor. This provides a measurement of the amount of light transmitted through the bone.
Do bone samples remain in the experimental set?
Yes
No
End
Boiling time = 90sec
Repeat measurement until sample is boiled for a total of 90 seconds
Boiling time < 90 sec
Remove the sample and dry it.
Place the bone sample in a pot of boiling water for 30 seconds.

Data Results Analysis



1-5: “Thin” samples, average thickness of 4.32 mm
6-12: “Thick” samples, average thickness of 10 mm

Data Results Analysis

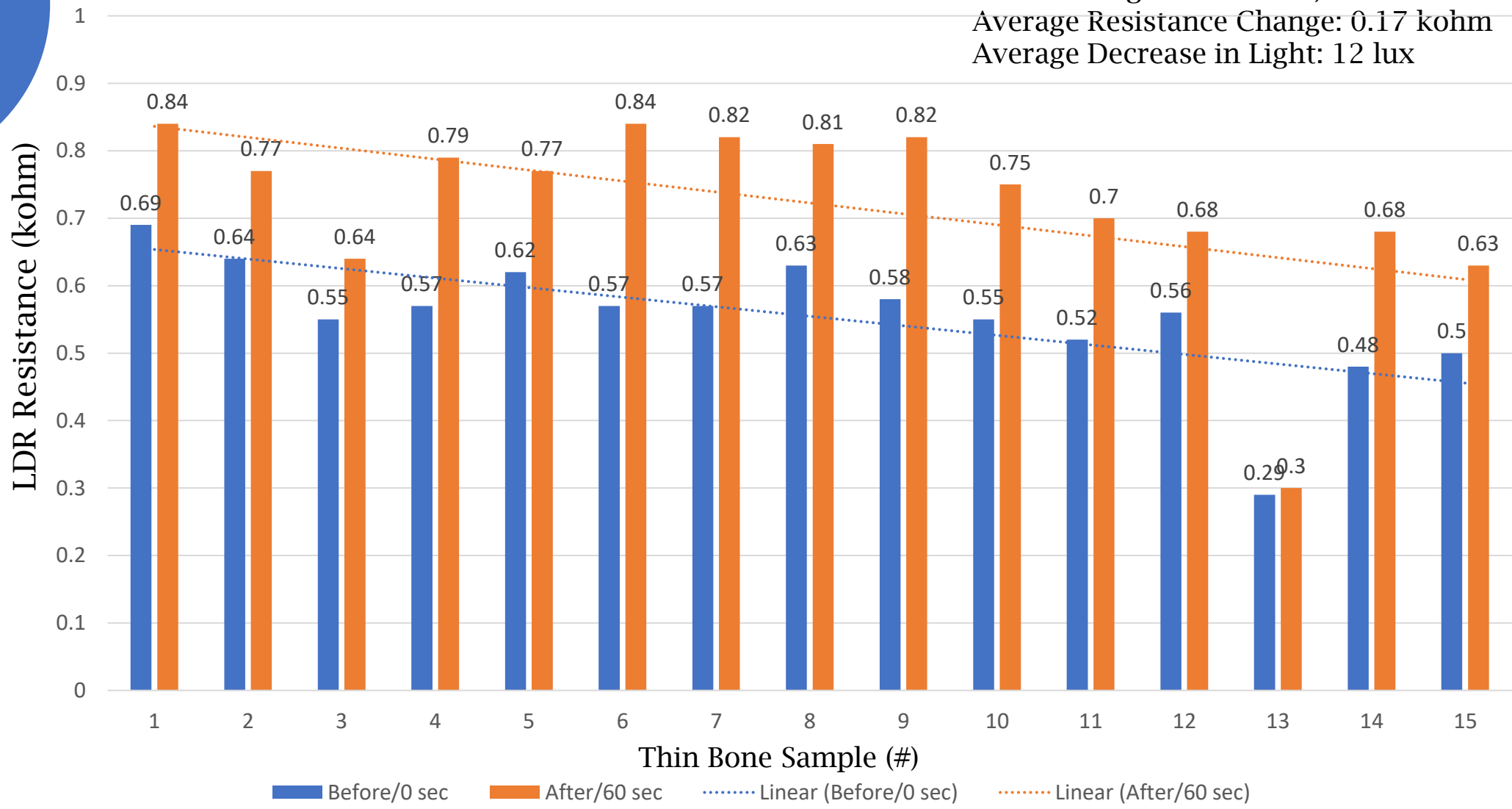


13-22: “Thin” samples, average thickness of 4.32 mm

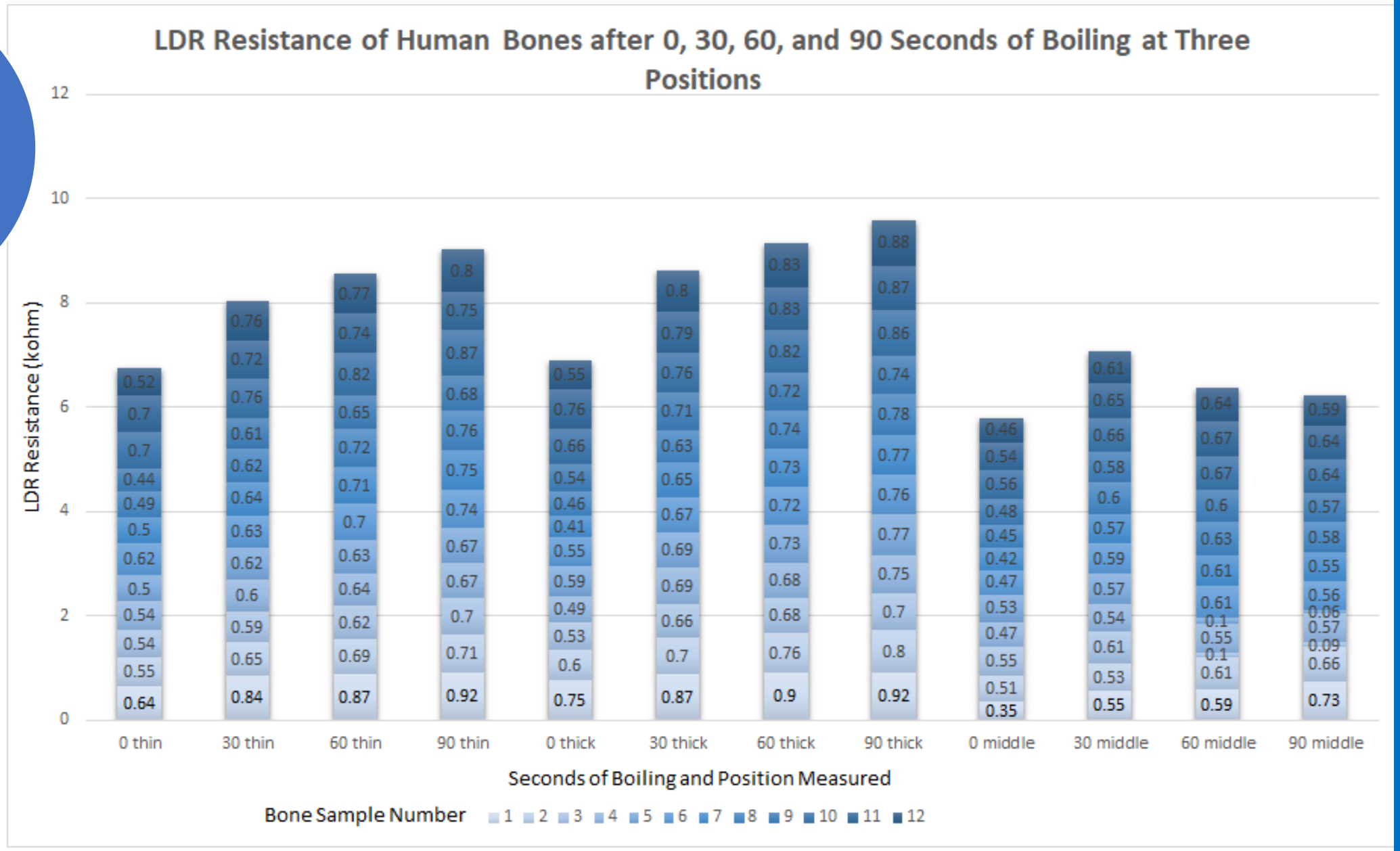
Data Results Analysis

Aggregate Change in LDR Resistance for all Thin Bovine Bone Samples

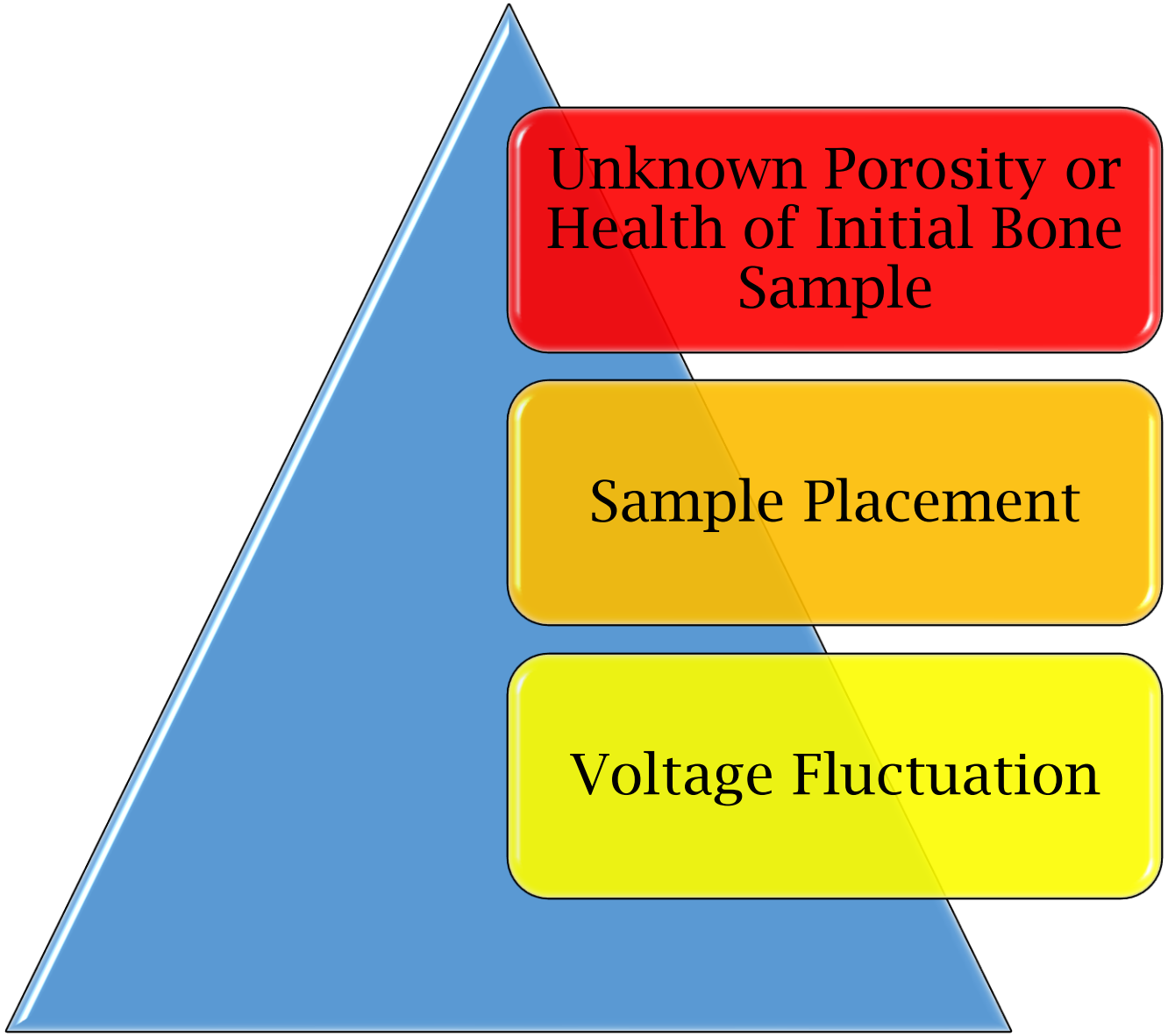
Before Average: 0.55 kohm, 20 lux
After Average: 0.72 kohm, 8 lux
Average Resistance Change: 0.17 kohm
Average Decrease in Light: 12 lux



Data Results Analysis



Error Analysis



Future Plans

Implement and Evaluate Ultrasound

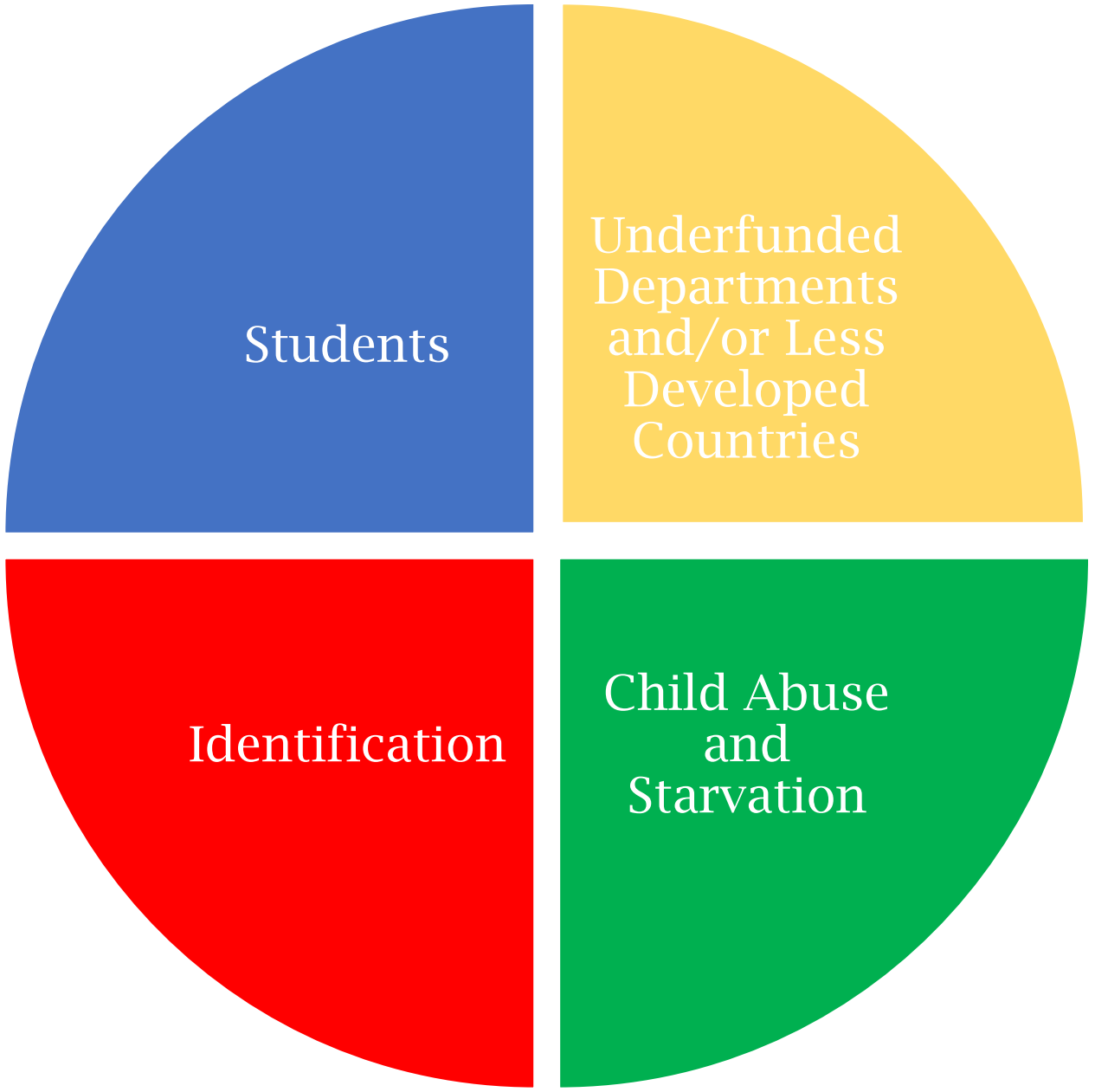
Implement and Evaluate Ultraviolet Light

Implement and Evaluate Infrared Light

Establish a Clinical Validation and Conversion to T-score

Non-Invasive, In-Vivo Bone Density Test for Determining and/or Monitoring Osteoporosis

Applications



Conclusion

Results

- Translucence: healthy bone samples > porous, as indicated by lower LDR resistance values
- Correlation between porosity and translucence between samples in the same category as well as across categories

Analysis

